

L_RAG: A Controlled Ablation Study of Vietnamese Legal Retrieval-Augmented Generation

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Abstract

Legal question answering must retrieve evidence that is relevant, complete in its statutory context, valid at the relevant time, and traceable to an authoritative source. We evaluate L_RAG, a Vietnamese legal retrieval-augmented generation pipeline that combines E5 dense retrieval, BM25, a knowledge-graph overlay, reciprocal-rank fusion, second-stage reranking, and citation-grounded generation. On a fixed 500-question benchmark (400 answerable and 100 unanswerable questions), we report five controlled ablation analyses grouped into four run families rather than a pooled factorial comparison: embedding text; dense, hybrid, and graph retrieval as one joint retrieval/graph run family; reranking within a shared candidate pool; and Base versus chain-of-thought (CoT) prompting. Metadata improves document-level success at depth 10 by 4.5 percentage points but reduces exact-chunk MRR. The tested BM25+RRF hybrid loses 5.98 Recall@10 points to dense retrieval; its recorded median total rises from 5.483 to 31.538 seconds under an accounting convention that includes amortized BM25 precomputation and is not online request latency. One-hop graph expansion loses 11.81–12.40 Recall@10 points, chiefly by replacing seed evidence. In a common 30-candidate pool, Jina reranking improves Recall@10 by 9.23 points and nDCG@10 by 11.85 points over RRF-only. Under fixed context, CoT decreases GPT-4o-mini Token F1 by 1.57 points. Retrieval is scored against gold chunk identifiers; lexical answer overlap, template-based answerability behavior, structural citation fields, and recorded latency are separate diagnostics. These are benchmark-bounded engineering findings: they do not establish legal correctness, citation entailment, or faithful reasoning.

1 Introduction

Legal information access is an evidence-sensitive task: a useful answer must identify the govern-

ing text, preserve the context in which a provision operates, and allow the reader to inspect its source. These requirements are especially important for Vietnamese law, where relevant rules are distributed across laws, decrees, decisions, and circulars connected by hierarchical, referential, and temporal relationships. This report studies L_RAG, a Vietnamese legal Retrieval-Augmented Generation (RAG) prototype that combines corpus structuring, dense and sparse retrieval, a legal knowledge graph, reranking, and citation-grounded answer generation. The central question is not whether adding more components makes a pipeline appear more sophisticated, but which components measurably improve evidence retrieval and answer behavior under controlled comparisons.

1.1 Background and Motivation

Keyword search remains valuable in legal research when a user already knows an article number, document title, or exact statutory phrase. In ordinary question answering, however, users often describe a situation in everyday Vietnamese rather than reproduce the terminology of an official instrument. Dense retrieval can bridge this lexical gap by matching semantic content, and large language models can turn retrieved passages into readable answers. Neither capability alone is sufficient for dependable legal assistance.

Legal documents are structured and interdependent rather than independent blocks of text. The meaning of a clause may depend on its article heading, parent document, neighboring provisions, referenced instruments, or effective date. A passage that is topically similar can therefore be incomplete, superseded, or disconnected from the rule needed to answer the question. Moreover, a fluent generated response can conceal missing evidence or attach a citation that does not support the adjacent claim. These risks motivate a pipeline in

which retrieval units preserve provenance, graph relations expose legal structure, and generation is constrained by inspectable evidence.

L_RAG provides a common implementation and evaluation setting for comparing these choices. In particular, it enables controlled tests of metadata-enriched embeddings, dense versus dense-sparse retrieval, one-hop graph expansion, second-stage rerankers, and base versus explicit reasoning prompts. This evidence-driven framing is necessary because a component that is intuitively appropriate for legal text may still reduce exact evidence coverage or impose an unacceptable latency cost in a fixed pipeline.

1.2 Problem Statement

Given a natural-language Vietnamese legal question q , a corpus of legal documents $\mathcal{D}$, and retrieval units $\mathcal{U}$ derived from those documents, the system must select a compact evidence set $E \subset \mathcal{U}$ and generate an answer a whose material claims can be traced through chunk, provision, and document identifiers. A legal time or scope constraint can further restrict applicability when the corresponding overlay is explicitly supplied. The task is difficult because four failure modes interact:

1. **Lexical mismatch.** A user’s vocabulary may differ from the formal wording of the applicable provision, which weakens exact keyword matching.
2. **Structural fragmentation.** Chunking can separate a clause from its article header, parent instrument, definitions, exceptions, or adjacent provisions, causing retrieval of locally relevant but incomplete evidence.
3. **Relationship and validity blindness.** Semantic similarity alone does not represent citation, amendment, replacement, authority, or effective-date relationships between instruments.
4. **Ranking and generation error.** Relevant evidence may be present in a candidate pool but ranked outside the final context window; the generator may then omit support, refuse an answer unnecessarily, or produce an unsupported statement.

These failures have practical consequences: the system may cite the wrong instrument, overlook a controlling provision, present outdated material,

or produce an answer that cannot be audited. The research problem is therefore to determine how semantic, lexical, structural, and generative components can be combined while measuring both evidence quality and operational cost. In this report, success is evaluated through benchmark-bounded retrieval, answer overlap, answer/refuse behavior, structural citation, completion, and latency diagnostics; it is not treated as certification of legal correctness.

1.3 Project Objectives

The project pursues the following measurable objectives, each linked to an implemented module and a report artifact:

1. **Build traceable retrieval units from Vietnamese legal data.** Normalize documents into provisions and chunks with stable document, provision, and chunk identifiers and retained metadata. Corpus scale and graph integrity are reported in Sections 3 and 3.9.
2. **Implement comparable evidence-retrieval paths.** Provide dense E5 retrieval, Vietnamese-tokenized BM25 retrieval, Reciprocal Rank Fusion (RRF), and graph-based filtering or expansion behind a shared pipeline. Their behavior is measured using Recall@ k , Hit@ k , MRR@ k , nDCG@ k , precision, and latency in Sections 5 and 6.
3. **Model legal structure and inter-document relations.** Build document, provision, and chunk nodes with containment, reading-order, and cross-document edges, while retaining provenance and verification gates. The schema and traversal role are specified in Sections 4 and 4.4.
4. **Improve final-context ranking under a fixed evidence ceiling.** Compare RRF-only ordering with mMiniLM, Qwen3, BGE, and Jina rerankers over the same 30-candidate pool and measure their ability to retain and promote gold evidence into the top 10, together with reranking latency.
5. **Support grounded generation and explicit abstention.** Construct answers from the selected context, expose source markers, and instruct the generator to refuse when evidence is insufficient. Generated outputs are assessed

with lexical-overlap, template-based answerability, citation-structure, completion, and latency diagnostics, with their limitations stated explicitly.

6. **Create a reproducible evaluation basis.** Use a frozen 500-question benchmark—400 answerable and 100 unanswerable cases—with evidence identifiers and saved run artifacts. Paired comparisons and bootstrap confidence intervals isolate the tested component within each ablation family.

1.4 Scope of the System

The implemented scope covers the offline preparation of Vietnamese legal documents, structural parsing into provisions and chunks, dense and sparse index construction, knowledge-graph construction, candidate retrieval and fusion, graph-assisted context selection, second-stage reranking, prompt construction, citation-grounded generation, and offline evaluation. The experimental scope is limited to the available corpus and frozen benchmark, the recorded index and graph versions, the tested retrieval budgets, and the models and prompts documented in Section 5. The reported graph ablation tests bounded structural expansion with broad validity filtering; it does not test the separate query-time temporal-validity or authority overlay.

The project does not claim exhaustive coverage of Vietnamese law, real-time synchronization with authoritative legal databases, resolution of every amendment or relationship-direction ambiguity, or validation of the generated answers by legal experts. The benchmark is partly synthetic and contains relatively small multi-hop and cross-document slices. Automatic overlap and citation-marker metrics do not establish legal correctness, completeness, or citation entailment. L_RAG is therefore a research prototype and evaluation platform, not a source of legal advice or an autonomous decision-making system. Any deployment affecting rights, obligations, or remedies must verify the cited text and its current legal status and retain qualified human review.

1.5 Main Contributions

This work makes five main contributions:

1. A processed Vietnamese legal corpus with document–provision–chunk provenance, comprising 151,624 retained document

records, 1,386,267 provisions, and 1,513,376 retrieval chunks.

2. A typed legal graph containing structural reading-order links and 883,256 retained cross-document relationship records plus 19,763 external stubs, together with quarantine records, a verified-edge view, temporal and authority overlay interfaces, and bounded context expansion. The overlay interfaces are documented system capabilities, not effects tested by the reported graph ablation.
3. A modular end-to-end pipeline that integrates multilingual dense retrieval, Vietnamese BM25, RRF, four second-stage rerankers, and evidence-constrained answer generation.
4. A 500-question Vietnamese legal QA benchmark with answerability, category, answer type, reference answer, and document-, provision-, and chunk-level evidence provenance.
5. Five controlled analyses grouped into four run families, separating component value from architectural intent. The experiments show that metadata primarily improves document-level identification, the tested hybrid and graph settings trade away exact evidence coverage, reranking provides the clearest retrieval gains, and explicit reasoning prompts do not automatically improve generated-answer overlap.

1.6 Report Organization

The report follows a Problem → Data → Methodology → Evaluation → Discussion progression. Section 2 positions the problem within RAG, legal IR, hybrid retrieval, and graph-augmented retrieval. Section 3 describes the processed corpus and QA benchmark, while Section 4 presents the L_RAG architecture and methodology. Section 5 defines the implementation, metrics, controls, and reproducibility protocol. Section 6 presents the controlled ablations and their results. Section 7 interprets the findings and states the technical and ethical boundaries, and Section 8 summarizes the evidence and identifies future work.

2 Background and Related Work

This section positions L_RAG at the intersection of retrieval-augmented generation, legal informa-

tion retrieval, hybrid ranking, knowledge-graph retrieval, reranking, and citation-grounded generation. It first establishes the standard RAG workflow, then identifies the legal properties that ordinary text retrieval does not capture. The purpose is not to claim that one paradigm solves legal QA by itself, but to make explicit which prior ideas motivate each component and which evaluation check tests the corresponding design assumption. Table 1 provides the design-to-evidence roadmap before the individual research strands are discussed.

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) supplements a language model’s parameters with documents retrieved at query time. A standard pipeline contains five logical stages: (1) represent the query and retrieve candidates from an indexed corpus; (2) select, order, and possibly expand a bounded context; (3) serialize that context with provenance into a prompt; (4) generate an answer conditioned on the prompt; and (5) check whether the answer is supported by, and correctly attributes, the supplied evidence. The external context can reduce dependence on stale parametric knowledge and gives a user sources to inspect, but it does not guarantee that the selected passages are sufficient or that the generated claims follow from them.

The basic RAG abstraction is useful for L_RAG because it separates three possible failure locations: the retriever can miss the governing passage, the context builder can omit its parent or exception, and the generator can fail to use or cite the evidence. Advanced methods such as Self-RAG (Asai et al., 2024) and Corrective RAG (Yan et al., 2024) add self-critique or retrieval-quality assessment and correction. They motivate separating retrieval quality from answer behavior, but they do not by themselves encode the formal hierarchy, amendment history, or temporal validity of Vietnamese legal instruments.

2.2 Legal Information Retrieval

Legal Information Retrieval (Legal IR) is a specialized area of information retrieval and legal informatics (Locke and Zuccon, 2022). Legal texts differ from open-domain documents in ways that change both indexing and evaluation:

1. **Hierarchical structure.** An instrument is organized into parts, chapters, articles, clauses,

and items. The meaning of a chunk can depend on its parent heading or a neighboring exception.

2. **Precision requirements.** A missing qualifier, number, or negation can change the interpretation of a rule even when the retrieved paragraph is topically correct.
3. **Authority and validity.** The Constitution, temporal validity of Vietnamese legal instruments.

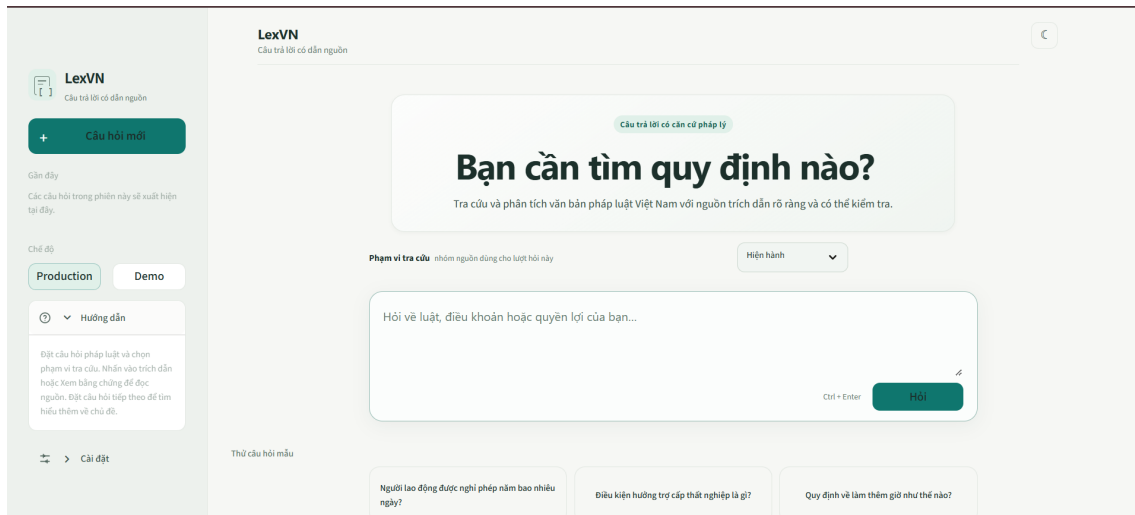


Figure 4: Initial LexVN interface for submitting a Vietnamese legal question. The sidebar exposes session controls, Production/Demo mode, guidance, and retrieval settings, while the main panel provides the query input and suggested questions.

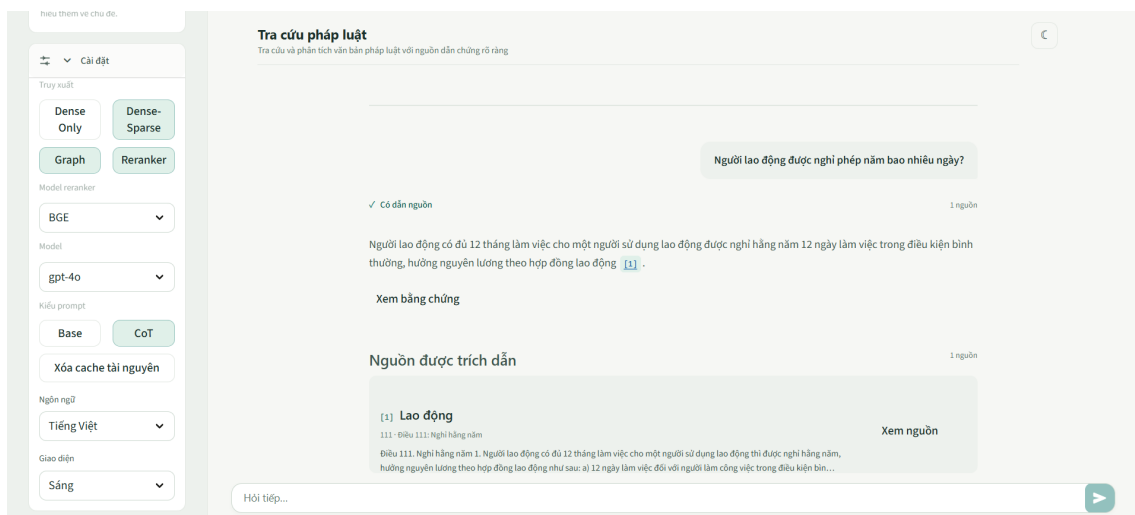


Figure 5: Grounded-answer view for an annual-leave question. The generated answer includes an inline citation, while the supporting source is displayed below for direct inspection.

to complete the purchase of fire-prevention equipment:

“Có thời hạn cụ thể để cơ sở hoàn thành việc mua sắm trang thiết bị phòng cháy, chữa cháy không?”

When the available evidence does not support a reliable answer, the interface displays the state “Chưa đủ bằng chứng” rather than presenting an unsupported legal conclusion. No citation is attached to this response because no supporting passage is available for the requested claim.

Together, these interface states expose the main user-facing L_{RAG} workflow: submitting a legal question, receiving a grounded response, in-

specting the cited passage and its provenance, and receiving an explicit abstention when sufficient evidence is unavailable. The interface therefore makes the distinction between answer generation, evidence inspection, and insufficient-evidence handling visible to the end user.

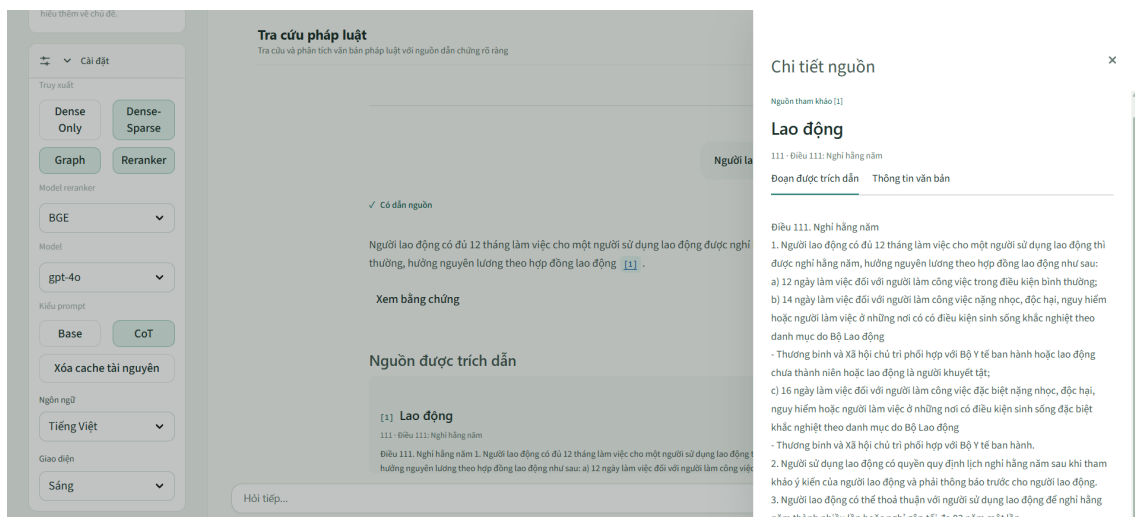


Figure 6: Extracted legal passage associated with the cited source. The panel exposes the text from Article 111 on annual leave that supports the generated answer.

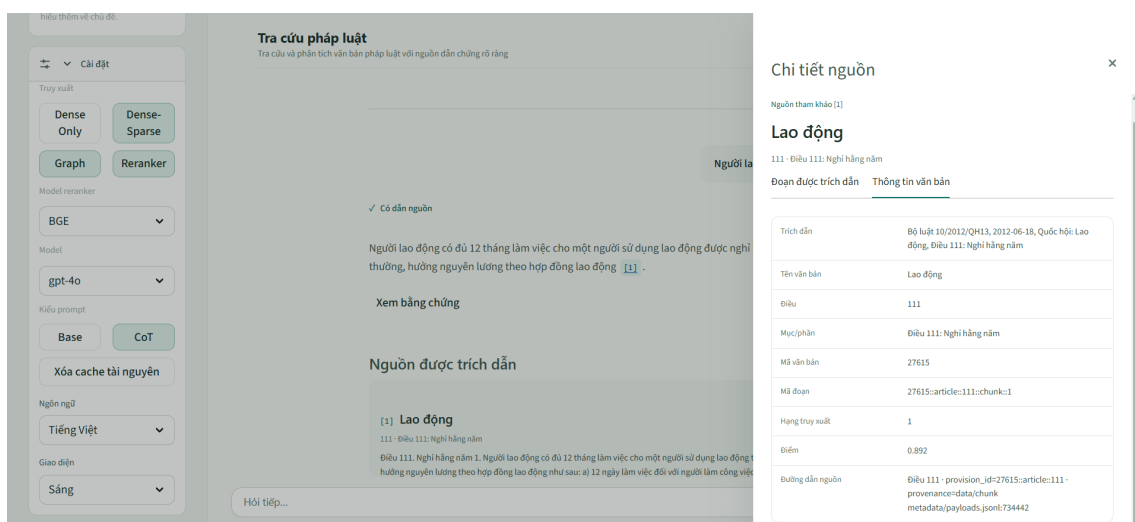


Figure 7: Document and provenance metadata for the cited source, including the document title, provision, chunk identifier, retrieval rank, score, and source path.

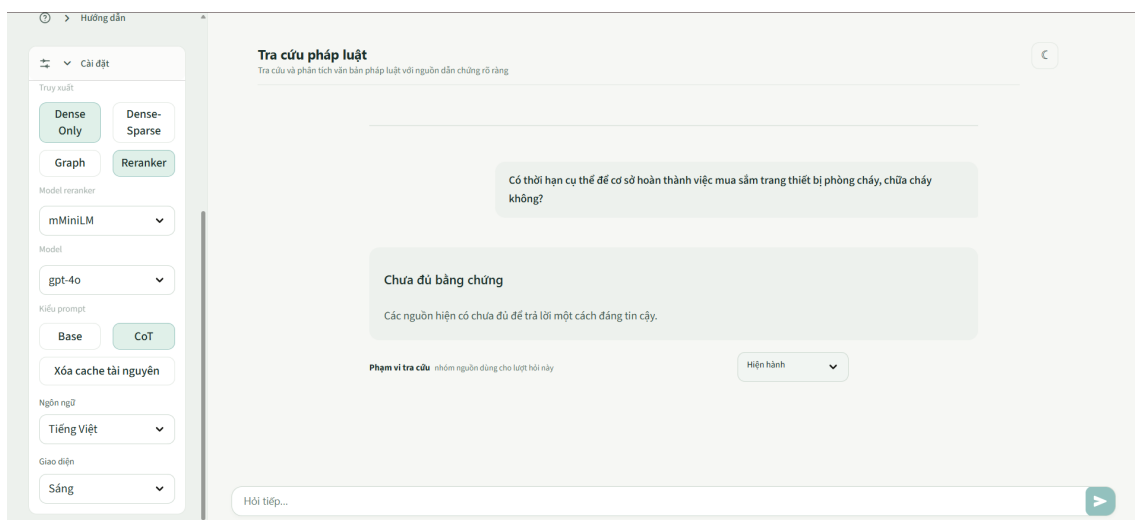


Figure 8: Insufficient-evidence state. When the available sources do not support the requested legal information, the interface explicitly indicates that the evidence is insufficient instead of generating an unsupported answer.